Bowl Reflector Optical Coupling | Technical Report

Mathematical Modeling · Python Algorithms · Zemax Simulation · Data Analysis · Debugging

Research simulation (October 2026)

Scope: simulation and design only; no experimental fabrication or measured fiber-mode coupling is claimed.

1. Executive Summary

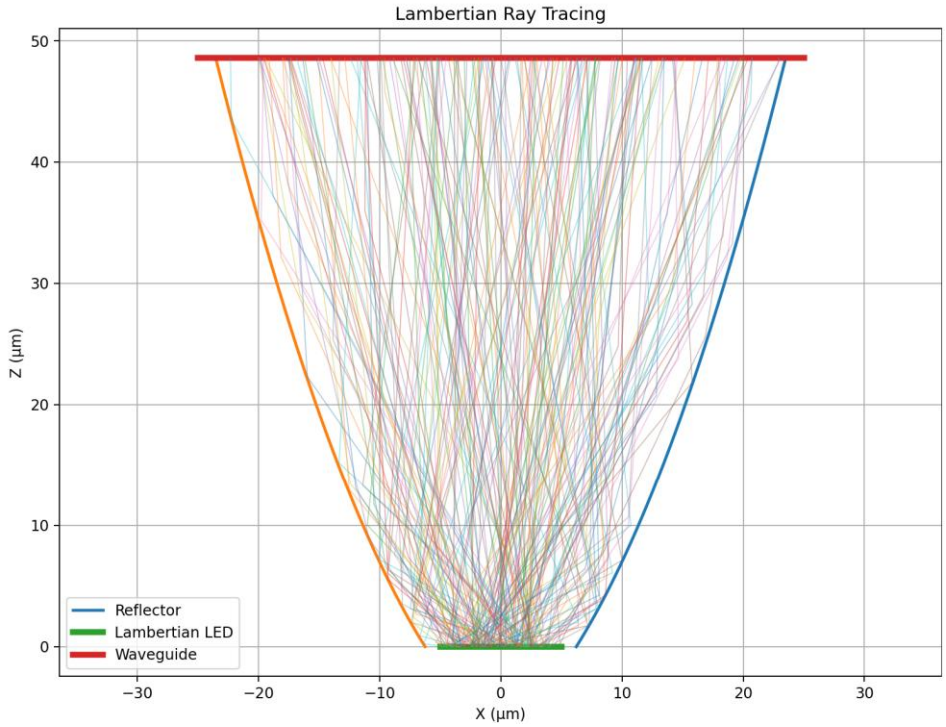
This project implements an axisymmetric bowl-reflector design workflow using Python, exports a sampled surface for Zemax-oriented evaluation, and compares geometric Monte Carlo power collection with non-sequential detector results. The goal is to demonstrate numerical modeling, software implementation, optical analysis, and verification discipline.

Metric	Recorded result
Python Simulation :TOTAL coupling efficiency	95.6774%
Estimated geometric power loss	0.1919 dB
Ray samples	100,000
Assumed reflector reflectivity	0.95 per reflection
Zemax unfiltered detector power	0.79392–0.93905 W

Important: Python power collection is NOT guided-mode coupling efficiency. Zemax detector power is NOT a normalized efficiency until incident power and detector definition are verified.

2. Optical Architecture and Model Scope

Optical path: circular Micro-LED source → direct rays / specular bowl-reflector rays → receiving plane representing a waveguide/fiber aperture. A position-and-angle acceptance test is used. Waveguide modal overlap, polarization, coherence, scattering, and measured insertion loss are not simulated.



Original simulation figure

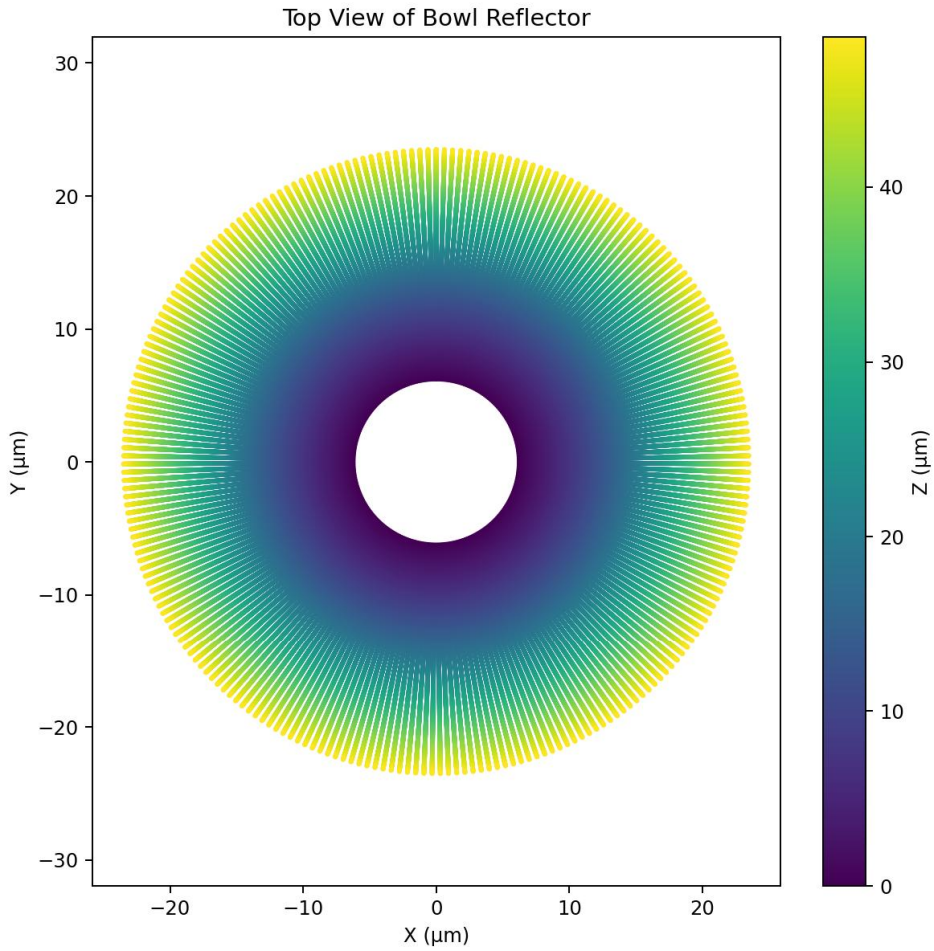
3. Mathematical Modeling

3.1 Geometric design

The code defines wl_design=10 µm, ww_design=50 µm, alpha_deg=30°, dimensional margins, and reflector_depth_ratio=0.90. The initial reflector diameter is wr_design(0)=ww_design−dw_r−dw_w−2dr_rw=47 µm; initial radius is 23.5 µm. A design-height expression in the code gives h_rl_design=53.9615 µm and plotted reflector depth h_r=48.5654 µm.

3.2 Numerical integration and slope bounds

The implemented reflector ODE is dr/dz = Sr_design(z,r), where Sr_design is selected from code-defined lower/upper slope bounds. SciPy solve_ivp integrates the profile from z=0 to z=−h_r, sampled at 800 points. The resulting axisymmetric surface is revolved and discretized for visualization/export. This is a numerical design rule encoded by the project, not an independently verified analytical optimum.



Reflector geometry

3.3 Ray and power model

Monte Carlo rays are generated from the circular source. Each ray may be accepted directly, reflected specularly, or lost. Reflectivity is modeled as 0.95 per bounce, with at most five reflections. Let P_in be total launched ray power, P_dir accepted direct power, and P_ref accepted reflected power. η_geo=(P_dir+P_ref)/P_in; L_geo=−10 log10(η_geo) dB. These symbols are explanatory definitions for this report, not new variables from an external paper.

4. Python Algorithm and Implementation

Stage	Implementation
Parameter configuration	Geometry, margins, angle, samples, seed
Constraint checks	Slope and source-enclosure checks
ODE integration	scipy.integrate.solve_ivp
Surface interpolation	scipy.interpolate.interp1d
Intersection solving	scipy.optimize.brentq and geometry checks
Ray tracing	Monte Carlo directions, surface normals, specular reflection
Power accounting	Direct / reflected / lost power
Export	201 × 201 sampled .GRD surface

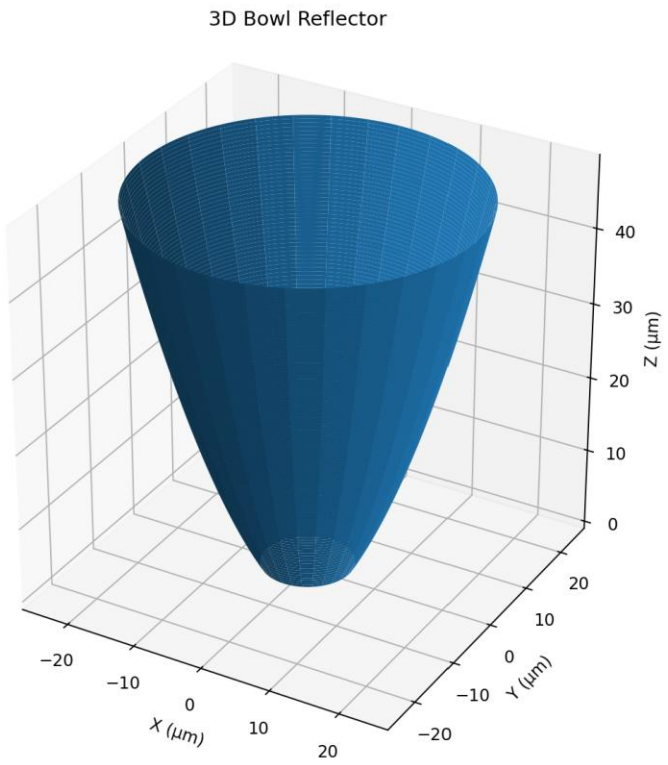
Reproducibility: the code uses NumPy RNG seed 42 and 100,000 rays. The printed lambertian_exponent=1.0 is not currently wired into the direction sampler, so the source distribution should be checked before making general Lambertian claims.

5. Simulation Results

5.1 Python ray tracing

Category	Count	Power fraction
Direct accepted	18,745	18.7450%
Reflected accepted	81,240	76.9324%
Lost	15	—
Total modeled collection	100,000 launched	95.6774%

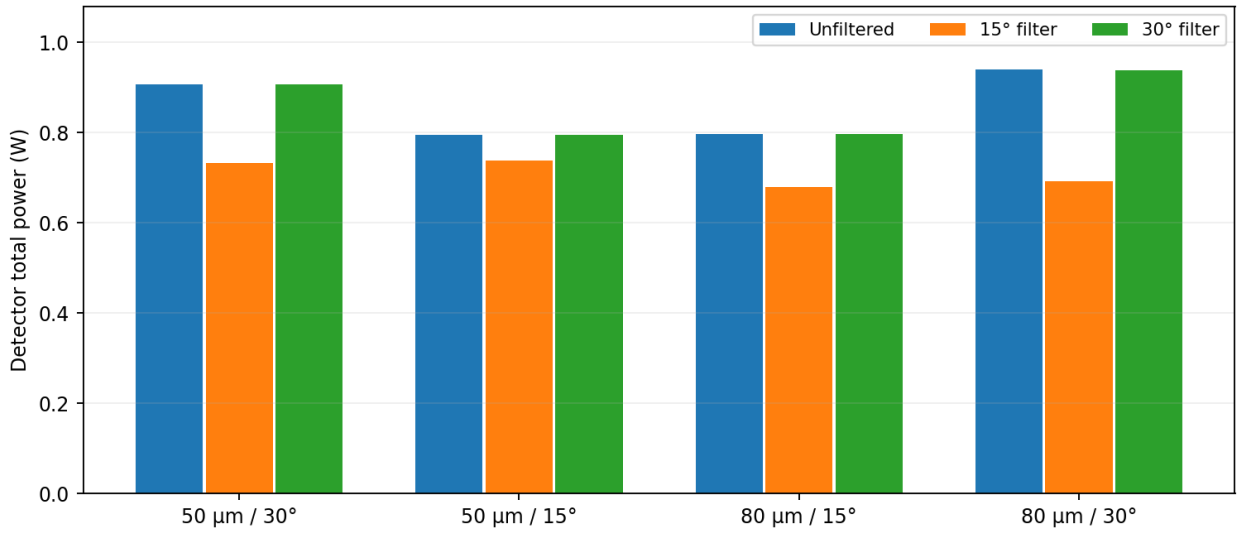
The 99,985 accepted-ray count must not be confused with 95.6774% power collection: reflected rays lose modeled power at each bounce. The reported power loss is 0.1919 dB.



Simulation output

5.2 Zemax detector comparison

Core / design angle	Unfiltered (W)	15° filter (W)	30° filter (W)
50 µm / 30°	0.90639	0.73247	0.90639
50 µm / 15°	0.79392	0.73792	0.79392
80 µm / 15°	0.79683	0.67883	0.79682
80 µm / 30°	0.93905	0.69176	0.93851



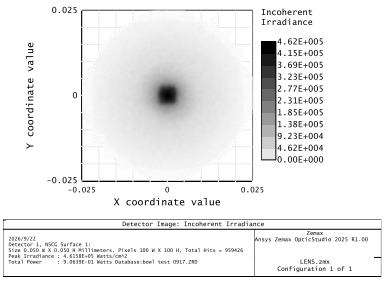
The 15°/30° detector filters use X_NGT(1,0.965926) and X_NGT(1,0.866), respectively. These filters are distinct from alpha_deg, a geometry/model design setting.

6. Data Analysis and Interpretation

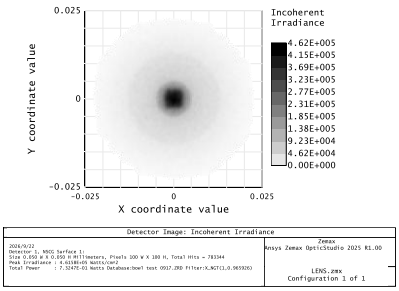
For a 50 µm receiver, changing the design angle from 15° to 30° changes unfiltered detector power from 0.79392 W to 0.90639 W (+0.11247 W). For an 80 µm receiver, it changes from 0.79683 W to 0.93905 W (+0.14222 W). Under 30° design, the 80 µm receiver yields 0.03266 W more than the 50 µm case. These are differences between recorded configurations, not controlled causal estimates unless all other settings are verified equal.

7. Zemax Simulation

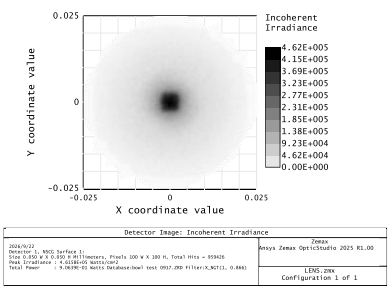
(1)50 µm / 30° :



Unfiltered (W)

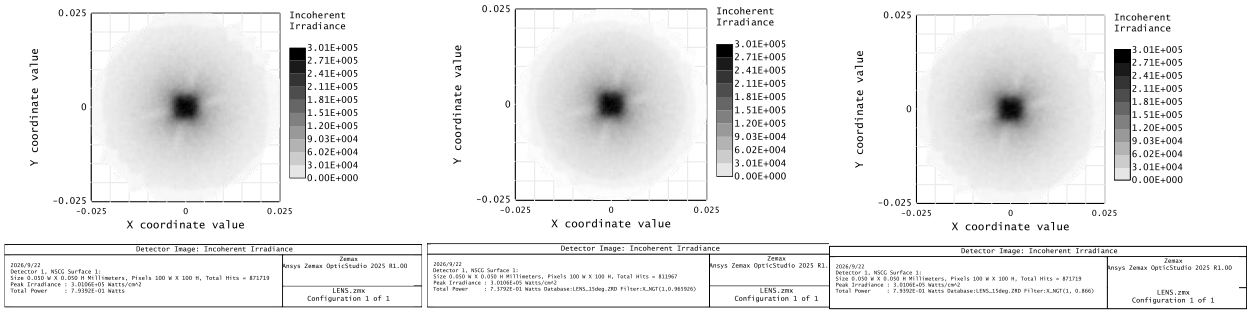


15° filter (W)



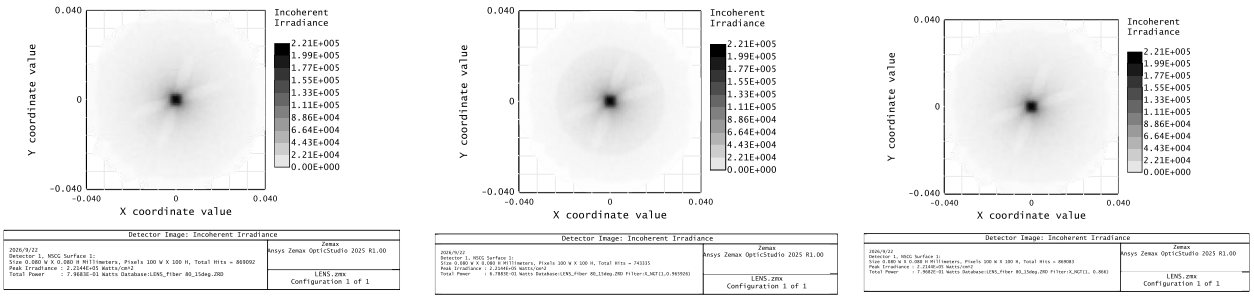
30° filter (W)

(2)50 μm / 15° :



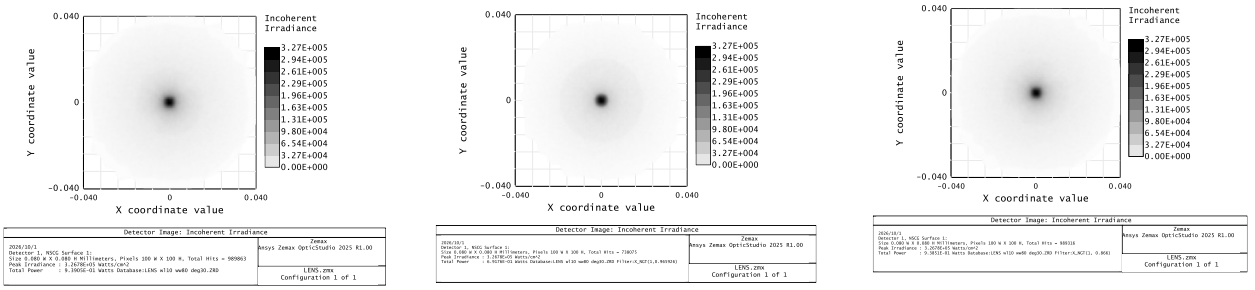
Unfiltered (W) 15° filter (W) 30° filter (W)

(3)80 μm / 15° :



Unfiltered (W) 15° filter (W) 30° filter (W)

(4)80 μm / 30° :



Unfiltered (W) 15° filter (W) 30° filter (W)

8. Conclusions and Engineering Skills

Demonstrated: translating geometry and physical assumptions into an ODE-based numerical solver; implementing ray/surface intersection and reflection logic; exporting structured simulation geometry; analyzing detector outputs with explicit filters; and identifying reproducibility/validation gaps. Next steps are standardized test configurations, automated result tables, convergence checks, and optical-mode coupling analysis.

Appendix | Traceability

Source code: main.py in Bowl-Reflector-Simulation-with-Zemax.zip.